Quantification: 543_z13
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	11
interior lacuna count	11
frame-edge lacuna count	0

Interior lacunae only: mean, median and sample SD (n = 11)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	1862	1736	823.012	px^2
major_axis_length_px	lacuna shape	78.3615	74.0218	22.3719	px
minor_axis_length_px	lacuna shape	30.8137	31.6227	6.9101	px
aspect_ratio	lacuna shape	2.5873	2.6086	0.6752	unitless
eccentricity	lacuna shape	0.9006	0.9236	0.0676	unitless
solidity	lacuna shape	0.9037	0.9231	0.0388	unitless
roots_count	headline	9.2727	10	3.3791	count
ring_length_r30_px	headline	377.182	392	59.543	px
ring_length_r60_px	reported	1028.18	1013	131.706	px
owned_length_px	ownership-dependent	3494.14	3322.18	1358.73	px
edge_count	ownership-dependent	158.727	132	63.3642	count
mean_edge_length_px	ownership-dependent	22.2109	21.9425	2.262	px
perimeter_px	normalised	207.479	205.037	54.0901	px
ring_area_r30_px2	normalised	8447.82	8219	1466.9	px^2
ring_area_r60_px2	normalised	21999.7	21794	3275.21	px^2
in_frame_fraction_r30	normalised	1	1	0	unitless
in_frame_fraction_r60	normalised	1	1	0	unitless
ring_density_r30	normalised	0.0452897	0.0440449	0.00782109	px^-1
ring_density_r60	normalised	0.0471902	0.0478606	0.00578824	px^-1
roots_per_100px_perimeter	normalised	4.4972	4.0874	1.483	per 100 px
ring_attached_length_r30_px	network v2	337.546	338	77.0485	px
ring_attached_length_r60_px	network v2	825.818	804	207.333	px
ring_length_w_r30_px	network v2	427.512	435.321	65.4763	px
ring_length_w_r60_px	network v2	1164.18	1126.77	149.807	px
sholl_crossings_r10	network v2	9	9	2.6458	count
sholl_crossings_r20	network v2	12.6364	13	2.4196	count
sholl_crossings_r30	network v2	13.8182	14	2.316	count
loop_count	loops, report only	17.1818	14	10.4768	count
loop_edge_share	loops, report only	0.4033	0.3865	0.0933	share
ring_width_mean_r30_px	width	7.3801	7.379	0.5054	px

Per field
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measure	value	units
analysed_area_px2	1.02809e+06	px^2
total_skeleton_length_px	44246	px
canalicular_length_density_per_px	0.0430369	px^-1
junction_count	1062	count
junction_density_per_px2	0.00103298	px^-2
lacuna_count	11	count

lacunae_per_px2	1.07e-05	px^-2
skeleton_component_count	203	count
mean_component_length_px	217.961	px
median_component_length_px	22	px
field_length_density_w_per_px	0.0488935	px^-1
field_density_without_flagged_per_px	0.0430369	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	256	px
unattached_component_count	13	count
canalicular_length_density_attached_per_px	0.0430369	px^-1
field_length_density_with_unattached_per_px	0.0432859	px^-1
excluded_area_px2	0	px^2
analysed_field_area_px2	1.04858e+06	px^2
loop_count	231	count
loop_edge_share	0.412	share
edge_count	2192	count

Canalicular width, the new measure

measure	value	units
width_mean_px	7.7853	px
width_median_px	8	px
width_p10_px	4.4721	px
width_p90_px	10	px
width_pixels_used	44246	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel.
Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the
dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the
pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every
thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	2019	79.1535	33.2822	2.3783	0.9073	0.9417	0.4378	97.9678	270.286	False
2	839	61.6665	17.863	3.4522	0.9571	0.906	0.068	131.349	828.439	False
3	1736	73.3356	33.3815	2.1969	0.8904	0.8374	0.2312	218.82	919.248	False
4	1138	46.8317	31.6227	1.481	0.7376	0.9252	0.396	263.104	340.498	False
5	2267	87.9083	33.2852	2.6411	0.9255	0.9473	0.0042	459.621	481.093	False
6	860	45.4745	26.4762	1.7176	0.813	0.854	0.07	475.221	748.086	False
7	2150	103.918	27.5311	3.7745	0.9643	0.9231	0.2711	546.673	873.734	False
8	3069	109.359	37.084	2.9489	0.9407	0.9099	0.049	642.67	188.33	False
9	1616	74.0218	28.3763	2.6086	0.9236	0.9245	-0.0477	731.536	630.238	False
10	1438	72.5092	25.5367	2.8394	0.9359	0.9236	0.0953	793.137	478.379	False
11	3350	107.799	44.5116	2.4218	0.9108	0.8475	-0.088	854.917	288.104	False

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	5	298	846	3539	163	21.7116	7.5162
2	11	407	1013	5340.9	237	22.5354	7.1326
3	13	372	923	1906.68	82	23.2521	6.7416
4	4	276	919	2095.26	112	18.7076	7.9621
5	10	392	1124	5814.77	265	21.9425	7.7742
6	5	396	989	2359.64	124	19.0294	7.3783

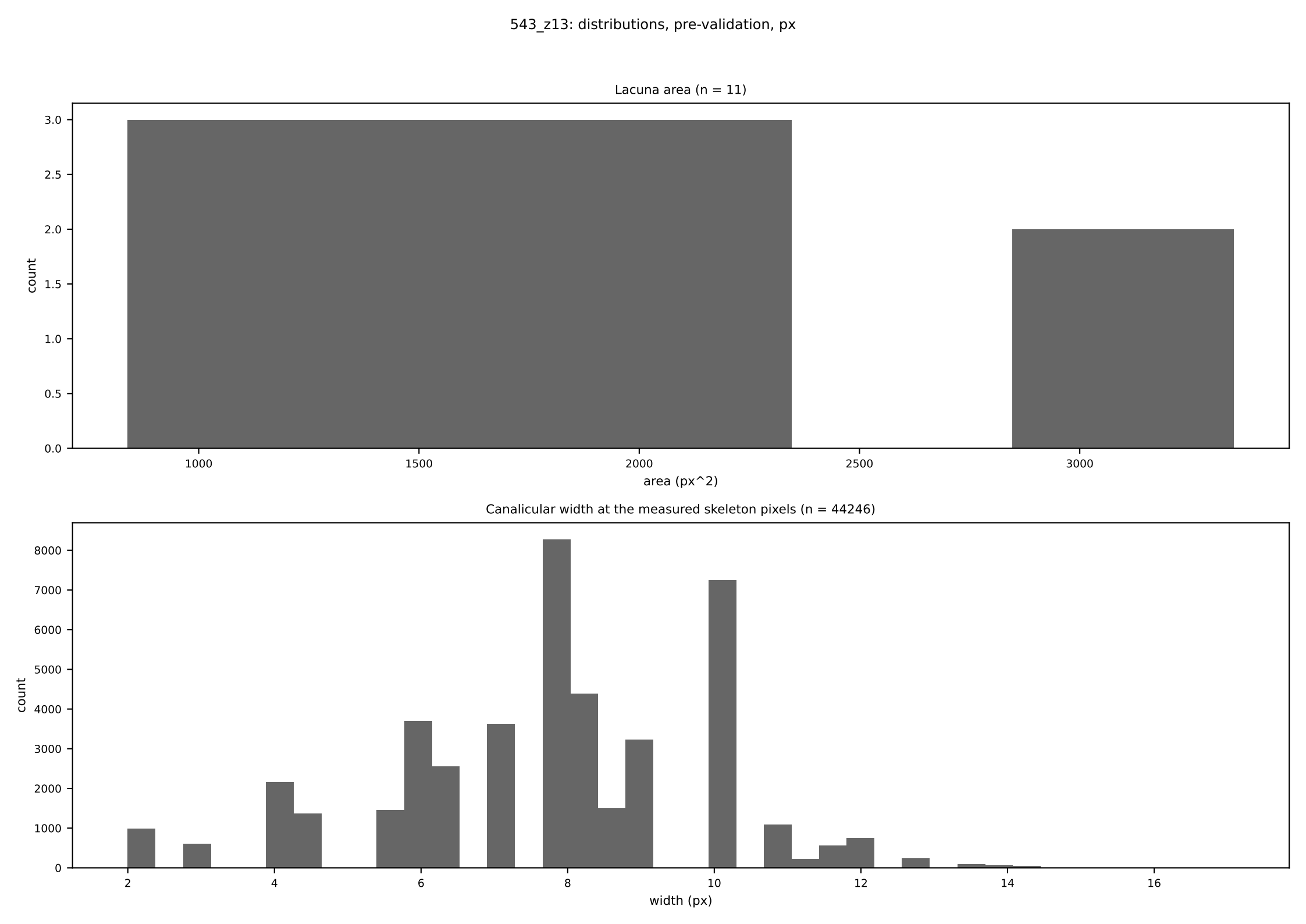
7	14	441	1170	4311.93	202	21.3462	6.6814
8	10	403	1080	3322.18	132	25.1681	7.8306
9	8	322	915	2315.54	91	25.4455	7.379
10	10	367	1037	2788.86	114	24.4637	6.7286
11	12	475	1294	4640.8	224	20.7179	8.0565

Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	205.037	8499	22530	1	1	0.035063	0.0375499	2.4386
2	151.297	6982	17707	1	1	0.0582927	0.057209	7.2705
3	205.037	8219	20045	1	1	0.045261	0.0460464	6.3403
4	139.681	6734	18999	1	1	0.040986	0.048371	2.8637
5	223.823	8900	23453	1	1	0.0440449	0.0479256	4.4678
6	141.338	6516	17817	1	1	0.0607735	0.0555088	3.5376
7	254.936	9810	24446	1	1	0.0449541	0.0478606	5.4916
8	278.593	10387	26403	1	1	0.0387985	0.0409044	3.5895
9	195.723	8083	21794	1	1	0.0398367	0.041984	4.0874
10	184.995	7835	21305	1	1	0.0468411	0.048674	5.4056
11	301.806	10961	27498	1	1	0.0433355	0.047058	3.9761

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	283	673	353.505	975.477	9	12	9
2	397	898	450.007	1126.77	9	13	15
3	320	601	422.505	1045.13	12	14	12
4	182	512	317.836	1037.64	4	9	14
5	338	928	435.321	1248.78	9	13	16
6	350	804	442.492	1108.98	6	13	15
7	418	914	501.375	1329.96	12	14	13
8	398	1055	462.818	1243.79	11	13	17
9	276	738	361.936	1042.26	8	8	13
10	300	730	411.978	1169.48	7	13	12
11	451	1231	542.86	1477.73	12	17	16


543_z13: distributions, pre-validation, px
Lacuna area (n = 11)
count
area (px^2)
Canalicular width at the measured skeleton pixels (n = 44246)
count
width (px)